CHALLENGES AND SOLUTIONS

Student Performance Analytics Dashboard

Introduction

During the development of the **Student Performance Analytics & Decision Support System**, several business, data engineering, and reporting challenges were encountered. These challenges were analyzed through business analysis techniques, and appropriate solutions were implemented across the data pipeline, including Python ETL, Microsoft SQL Server, and Power BI.

The following section summarizes the key challenges, their impact, root causes, and the solutions implemented.

Challenge 1: Designing a Realistic Educational Dataset

Problem

The initial synthetic dataset lacked realistic relationships between students, assessments, questions, and subjects. Performance scores showed limited variation, making the analytics unrealistic.

Impact

- Dashboards generated repetitive insights.
- Student performance bands were poorly distributed.
- Risk analysis lacked meaningful differentiation.

Solution

The dataset was redesigned to simulate a real academic environment by introducing multiple related datasets:

- Students
- Tests
- Student Responses
- Question Bank
- Assessment Scores
- Chapter Analytics
- Student Risk Reports

This improved the realism of the analytical outputs and produced more meaningful dashboard insights.

Challenge 2: Data Quality Issues During ETL

Problem

The raw CSV files contained duplicate records, missing values, inconsistent formatting, and invalid data that affected downstream analysis.

Impact

- Incorrect aggregations
- Inaccurate KPI calculations
- Reduced confidence in dashboard results

Solution

A Python ETL pipeline was developed to:

- Remove duplicate records
- Handle missing values
- Standardize data formats
- Validate business rules
- Generate clean analytical datasets

Only validated data was loaded into SQL Server.

Challenge 3: SQL Data Import Errors

Problem

While importing CSV files into Microsoft SQL Server, several columns exceeded the predefined field lengths, resulting in data truncation errors.

Example:

Long question options in the **Question Bank** exceeded the maximum length defined for NVARCHAR columns.

Impact

- Import failures
- Incomplete records
- Interrupted database loading process

Solution

The SQL table definitions were updated by increasing the size of text-based columns (e.g., question text and answer options). This ensured successful data import without loss of information.

Challenge 4: Data Validation Before Analytics

Problem

Raw data contained inconsistencies such as duplicate records, NULL values, invalid percentage values, and missing assessment references.

Impact

Poor data quality could have produced misleading dashboard insights.

Solution

A dedicated SQL Data Validation script was developed to verify:

- NULL values
- Duplicate records
- Percentage ranges (0–100)
- Marks obtained do not exceed maximum marks
- Students with valid assessment records
- Referential integrity between related tables

These validation queries ensured the reliability of the analytical dataset before reporting.

Challenge 5: Database Design and Data Organization

Problem

Managing multiple datasets directly in Power BI reduced maintainability and limited opportunities for data validation.

Impact

- Difficult debugging
- Limited scalability
- Repeated transformations

Solution

A relational database was created in Microsoft SQL Server.
The database organizes educational data into structured tables such as:

- Students
- Tests
- Responses
- Question Bank

SQL was used to centralize and manage the cleaned datasets before visualization.

Challenge 6: Business Query Development

Problem

Stakeholders required answers to business questions that were not directly available through raw datasets.
Examples included:

- Which students are at high academic risk?
- Which chapters have the lowest accuracy?
- Which subjects have the highest failure rate?

Impact

Decision-makers lacked actionable information.

Solution

Twenty-five business-oriented SQL queries were developed to answer stakeholder questions, including:

- Student performance summaries
- Risk analysis
- Subject performance
- Chapter performance
- Question difficulty analysis
- Pass percentage
- High-performing students
- Students requiring intervention

These queries validated business requirements before dashboard development.

Challenge 7: Power BI Data Modeling

Problem

Incorrect relationships initially created ambiguous filter paths and inconsistent calculations.

Impact

Visuals displayed incorrect aggregations.

Solution

The Power BI data model was redesigned using a clean star-schema approach with appropriate relationships based on:

- Student ID
- Test ID
- Question ID

Unnecessary relationships were removed to ensure accurate filtering and reporting.

Challenge 8: KPI Calculation Accuracy

Problem

Initial KPI calculations produced inconsistent totals because several metrics were implemented using calculated columns.

Impact

Dashboard indicators did not always match expected business values.

Solution

Business KPIs were rebuilt using optimized DAX measures and validated against SQL query results and source data.
Key KPIs validated included:

- Average Score
- Pass Percentage
- High Risk Students
- Students Needing Support
- Average Accuracy

Challenge 9: Student Performance Classification

Problem

Raw percentage scores alone did not provide sufficient insight into student performance.

Impact

Teachers found it difficult to prioritize interventions.

Solution

A performance band classification model was introduced.
Students were categorized as:

- Top Performer
- High Performer
- Average
- Needs Support
- At Risk

This simplified academic monitoring and reporting.

Challenge 10: Academic Risk Identification

Problem

School management required early identification of struggling students instead of relying only on final examination results.

Impact

Delayed interventions could negatively affect student outcomes.

Solution

A risk classification model was developed using:

- Average Percentage
- Performance Band
- Attendance Category

Students were categorized into:

- Low Risk
- Medium Risk
- High Risk

This enabled proactive academic support and intervention planning.

Challenge 11: Dashboard Design and User Experience

Problem

The initial dashboards contained excessive visuals, inconsistent formatting, and limited readability.

Impact

Users found it difficult to interpret key insights quickly.

Solution

The dashboards were redesigned using Business Intelligence best practices:

- Consistent layout
- Uniform color palette
- Standardized KPI cards
- Clear visual grouping
- Interactive filtering
- Intuitive page navigation

The final solution consists of four focused dashboards:

- Executive Performance Overview
- Individual Student Performance Analysis

- Risk Analysis Dashboard
- Subject & Chapter Performance Analysis

Overall Outcome

The challenges encountered throughout the project led to significant improvements in the overall solution.
The final system delivers:

- High-quality, validated educational data through Python ETL and SQL Server.
- Reliable business insights using SQL queries and DAX measures.
- Interactive Power BI dashboards for strategic and operational decision-making.
- A scalable analytics solution that supports academic performance monitoring, early risk identification, and curriculum improvement.